

Skewed cycle

X17 (2017) — English translation

A mole of an ideal monatomic gas participates in the cyclic process $1 - 2 - 3 - 4 - 1$, the graph of which is presented in the figure. The pressure at points 1 and 3 are equal. Temperatures $T_1 = 4T_0$; $T_2 = 8T_0$; $T_3 = 6T_0$; $T_4 = 3T_0$, where T_0 is a known value. The universal gas constant is R .

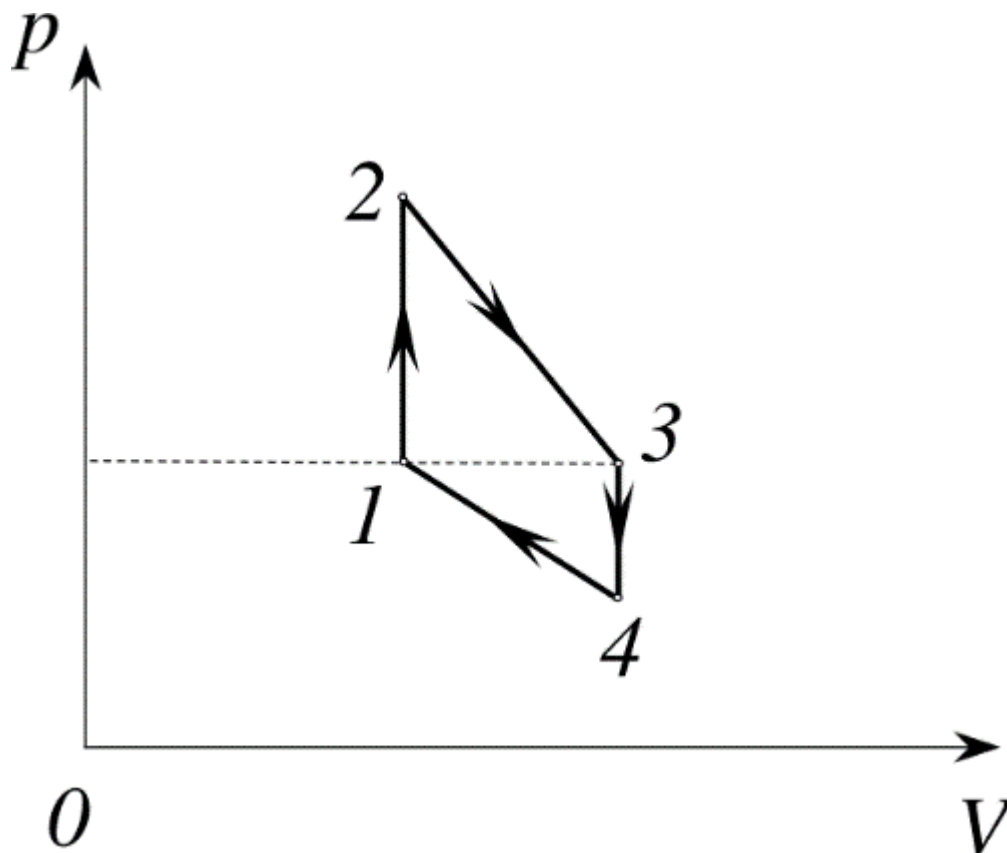


Fig. 1.

A1	Calculate the gas work A per cycle.	1.60
A2	Find the efficiency of cycle η .	6.40

Source: <https://pho.rs/p/303>